Expansive homeomorphisms on complexity quasi-metric spaces

A bridge between dynamical systems and computational complexity theory

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Abstract

The complexity quasi-metric of Schellekens is a topological framework in which the asymmetry of computational comparisons —"A is at most as fast as B" carrying different information than "B is at most as slow as A"—is built into the distance itself. This paper develops the theory of expansive homeomorphisms on the resulting space. The central result is that the scaling transformation $\psi_\alpha(f)(n) = \alpha f(n)$ is expansive on the complexity space $(\mathcal{C}, d_\mathcal{C})$ if and only if $\alpha \neq 1$. The δ-stable sets of this dynamics turn out to coincide with asymptotic complexity classes, giving a dynamical characterisation of objects familiar from complexity theory. We then show that ψ_α contracts $d^s_\mathcal{C}$-distances exactly by factor $1/\alpha$ per iterate, and we connect orbit separation to the classical time hierarchy theorem of Hartmanis and Stearns. Unstable sets, conjugate dynamics, and a no-compact-invariant-set theorem (showing that the standard Bowen entropy is trivially zero on compact invariant sets) are also worked out. Concrete algorithms and Python implementations accompany every proof, so each result can be checked computationally; SageMath snippets sit alongside the examples, and the full code is in the companion repository.

Keywords: Expansive homeomorphism; complexity quasi-metric; computational complexity; asymmetric topology; dynamical systems; stable and unstable sets; canonical coordinates

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"The study of iteration, the study of the behaviour of a transformation when it is repeated, is the fundamental problem of dynamics."

— Henri Poincaré

1 Introduction: where dynamics meets computation

1.1 A tale of two theories

This paper builds a bridge between two subjects that rarely meet: *dynamical systems*, which studies how a system evolves under iteration, and *computational complexity theory*, which studies the resources required to solve computational problems.

At first glance these two fields appear to have little in common. Dynamical systems theory asks: given a map $\psi\colon X \to X$, how do the orbits $\{x, \psi(x), \psi^2(x), \ldots\}$ behave as the number of iterations grows? Do nearby orbits stay close, or do they diverge? If they diverge, how quickly? Complexity theory, on the other hand, asks: given a computational problem, how do the resources required to solve it—time, memory, communication—grow with the input size n?

What links the two worlds is that complexity comparisons are asymmetric. Saying "algorithm A is at most as fast as algorithm B" does not say the same thing as "algorithm B is at most as slow as algorithm A." If $f(n) \le g(n)$ for all n (so that f is at least as fast as g), the "cost" of moving from f to g is zero—going to a slower algorithm is easy to simulate—but the cost of moving from g to f is positive, because shaving time off a running algorithm takes real insight.

This asymmetry is what a *quasi-metric* captures: a distance function q where $q(x, y)$ need not equal $q(y, x)$. Schellekens [10] introduced the *complexity quasi-metric* $d_{\mathcal{C}}$ on the space of functions $f\colon \mathbb{N} \to [1, \infty)$ and showed that its topological properties encode fundamental features of computational complexity. Romaguera and Schellekens [9] subsequently developed the quasi-metric structure of complexity spaces in depth.

On the dynamical side, *expansive homeomorphisms*—maps under which every pair of distinct points is eventually separated by more than some fixed threshold δ—are a central object of study, going back to Utz [12] and developed extensively by Bowen [2] and Reddy [8]. Recently, Olela Otafudu, Matladi, and Zweni [7] extended the theory of expansive homeomorphisms to quasi-metric spaces, opening the door to applications in asymmetric settings.

Our contribution is to walk through that door: to apply the theory of expansive homeomorphisms to the complexity quasi-metric space. Basic dynamical concepts—orbits, stable sets, hyperbolicity—turn out to have direct counterparts in complexity theory.

Related work. The complexity quasi-metric space was introduced by Schellekens [10], who established its basic topological properties and connections to denotational semantics. Romaguera and Schellekens [9] subsequently developed the quasi-metric structure in depth, proving completeness results and studying the Smyth completion. On the dynamical side, expansive homeomorphisms on metric spaces have a long history, beginning with Utz [12] and substantially advanced by Bowen [2], who connected expansiveness to topological entropy, and Reddy [8], who established canonical coordinate systems for expansive maps. The extension of expansive homeomorphisms to quasi-metric spaces was carried out by Olela Otafudu, Matladi, and Zweni [7], who proved that q-expansiveness and q^t-expansiveness are equivalent and developed the abstract theory of stable and unstable sets in the asymmetric setting. Künzi [5, 6] provided a comprehensive account of the topology of quasi-metric spaces, which underpins the present work.

What is new here is the *combination*: [7] develops the abstract theory without any specific quasi-metric space in mind, while [10, 9] study the complexity space without dynamical-systems tools. Bringing the two together gives dynamical concepts a computational meaning and gives complexity-theoretic distinctions a dynamical form.

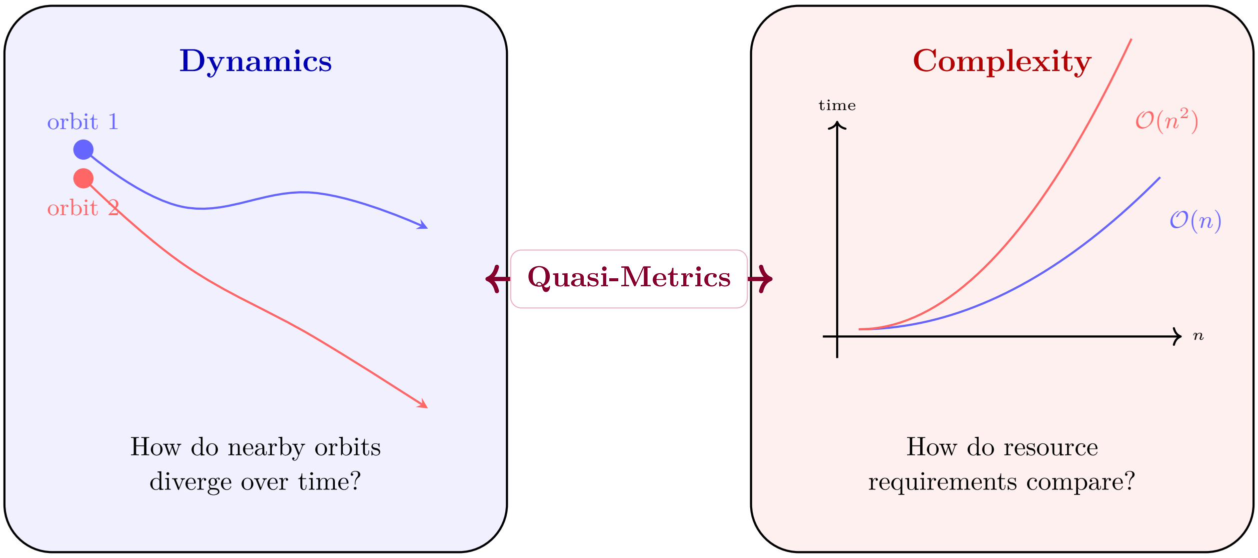

Figure 1: Two worlds connected by quasi-metrics. Dynamical systems study orbit divergence; complexity theory studies resource growth. The asymmetry inherent in both settings is encoded by the quasi-metric distance.

1.2 Why quasi-metrics?

Why quasi-metrics rather than ordinary metrics? The reason is short.

In a standard metric space (X, d), the symmetry axiom $d(x, y) = d(y, x)$ ensures that the cost of moving from x to y is the same as the cost of moving from y to x. This is natural in many geometric settings, but it is *unnatural* in computational settings. Consider two algorithms with running times $f(n) = n$ and $g(n) = n^2$. Given the faster algorithm f, one can trivially simulate the slower algorithm g by wasting time. But given the slower algorithm g, one cannot in general produce the faster algorithm f without effort. The "distance" from fast to slow should therefore be zero (or small), while the distance from slow to fast should be positive.

This is exactly what a quasi-metric provides. By dropping the symmetry axiom, quasi-metrics can encode directional costs, and the complexity quasi-metric $d_{\mathcal{C}}$ does precisely this: $d_{\mathcal{C}}(f, g) = 0$ whenever $f(n) \leq g(n)$ for all n (fast to slow is free), while $d_{\mathcal{C}}(g, f) > 0$ when g is genuinely slower (slow to fast is costly).

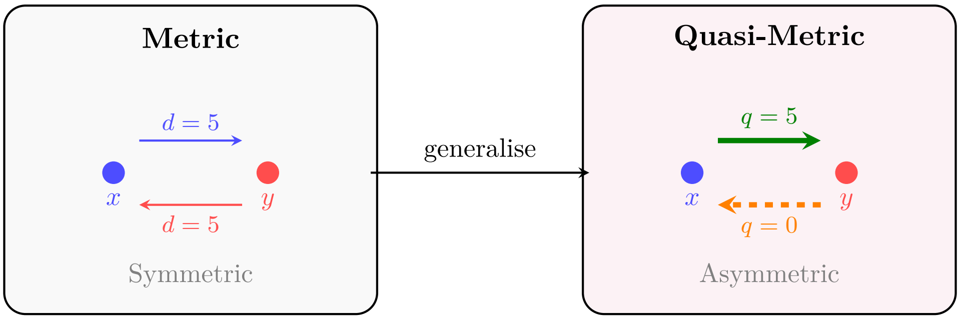

Figure 2: From metrics to quasi-metrics. Dropping symmetry allows the distance to encode directional information.

1.3 Overview of main results

Here is a preview of the main results; the reader may want to come back to this summary while working through the technical sections.

The first result characterises exactly when the scaling transformation is expansive.

5.9

The scaling map $\psi_\alpha(f)(n) = \alpha f(n)$ is expansive on the complexity space $(\mathcal{C}, d_\mathcal{C})$ if and only if $\alpha \neq 1$.

The second result gives the stable sets of this dynamics a clean complexity-theoretic interpretation.

6.2

For $\alpha > 1$, the δ-stable set of f under ψ_α coincides with the set $\{g : d_\mathcal{C}(f, g) \leq \delta\}$, which contains all functions g with $g(n) \geq f(n)$ for every n.

The third result is a precise form of hyperbolicity.

7.1

The scaling map ψ_α $(\alpha > 0)$ contracts distances exactly by factor $1/\alpha$ per forward iterate: the contraction rate is $\lambda = 1/\alpha$ and the constant is $C = 1$.

The fourth result connects the dynamical picture to a classical theorem in complexity theory.

8.1

If $f(n) \log f(n) = o(g(n))$, then the orbits of f and g under ψ_α separate in the symmetrized quasi-metric beyond every threshold.

Each of these results is proved in detail, with an accompanying algorithm and a Python implementation; all code lives in the companion repository.

1.4 Organisation

The paper is organised as follows. Section 2 recalls the basic theory of quasi-metric spaces, with examples and motivation. Section 3 introduces the complexity quasi-metric of Schellekens and establishes its fundamental properties, including several illustrative computations. Section 4 defines expansive homeomorphisms in the quasi-metric setting. Section 5 introduces the scaling transformation and proves our main expansiveness result. Section 6 develops the theory of stable and unstable sets. Section 7 establishes hyperbolicity. Section 8 connects orbit separation to the time hierarchy theorem. Section 9 discusses topological entropy estimates. Section 10 summarises the results and poses open problems. All Python implementations and SageMath verification scripts live in the companion repository (see the `code/` directory).

2 Quasi-metric spaces

We start by recalling the notion of a quasi-metric space. The theory has a long history, with major contributions by Künzi [5, 6], Cobzaş [3], and others. Notation and conventions follow Olela Otafudu et al. [7].

Definition 2.1 (Quasi-metric)**.** Let X be a non-empty set. A function $q\colon X \times X \to [0,\infty)$ is a *quasi-metric* on X if it satisfies the following three axioms for all $x, y, z \in X$:

(Q1) $q(x,x) = 0$;

(Q2) $q(x,z) \le q(x,y) + q(y,z)$ (triangle inequality);

(Q3) $q(x,y) = 0 = q(y,x) \;\Rightarrow\; x = y$ (T_0 separation).

The pair (X, q) is called a *quasi-metric space.*

Notice that the only axiom "missing" compared with a metric is symmetry: we do *not* require $q(x,y) = q(y,x)$. Dropping that single axiom changes the theory more than one might expect.

Every quasi-metric q gives rise to two natural companions.

Definition 2.2 (Conjugate and symmetrization)**.** Let (X, q) be a quasi-metric space. The *conjugate quasi-metric* is $q^t(x,y) := q(y,x)$. The *symmetrization* is $q^s(x,y) := \max\{q(x,y), q^t(x,y)\} = \max\{q(x,y), q(y,x)\}$.

It is straightforward to verify that q^t is again a quasi-metric and that q^s is a genuine metric on X. Thus every quasi-metric space carries a canonical metric, obtained by taking the "worst-case direction" of the asymmetric distance.

Example 2.3 (Standard quasi-metric on $\mathbb{R}$)**.** Define $u\colon \mathbb{R} \times \mathbb{R} \to [0,\infty)$ by

$$u(x,y) \;=\; (y-x)^+ \;=\; \max\{0,\, y-x\}.$$

Then u is a quasi-metric: (Q1) is immediate; (Q2) follows from $(z-x)^+ \le (y-x)^+ + (z-y)^+$; and (Q3) holds because $u(x,y) = 0 = u(y,x)$ gives $y \le x$ and $x \le y$, hence $x = y$.
The conjugate is $u^t(x,y) = (x-y)^+$, and the symmetrization is $u^s(x,y) = |x-y|$, the usual absolute-value metric.

The quasi-metric u has a vivid interpretation: *going uphill costs effort, while going downhill is free.*

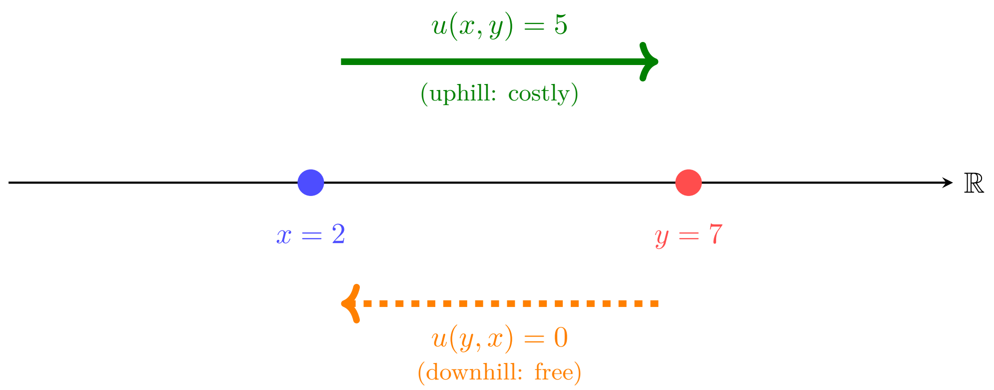

Figure 3: The standard quasi-metric on $\mathbb{R}$: going from $x = 2$ up to $y = 7$ costs $u(2, 7) = 5$, but going from $y = 7$ down to $x = 2$ is free: $u(7, 2) = 0$.

Example 2.4 (Weighted quasi-metric on $\mathbb{R}$)**.** For any weight $w > 0$, define $u_w(x, y) = w \cdot (y - x)^+$. This is again a quasi-metric, and it models the situation where the cost of going uphill is proportional to w. When $w = 1$ we recover the standard quasi-metric. The symmetrization is $u_w^s(x, y) = w|x - y|$.

Example 2.5 (Discrete quasi-metric)**.** On any set X, define

$$q_d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

This is both a metric and a quasi-metric (the symmetric case). It illustrates that every metric is automatically a quasi-metric.

Example 2.6 (Non-example: failing the triangle inequality)**.** On $\mathbb{R}$, define $\rho(x, y) = (y - x)^2$ if $y \geq x$ and $\rho(x, y) = 0$ if $y < x$. Then ρ satisfies (Q1) and (Q3), but it fails (Q2): take $x = 0$, $y = 2$, $z = 3$. Then $\rho(x, z) = 9$, while $\rho(x, y) + \rho(y, z) = 4 + 1 = 5 < 9$. Hence ρ is *not* a quasi-metric. This illustrates that the triangle inequality is a genuine restriction even in the asymmetric setting.

Example 2.7 (Asymmetric topologies on $\{a, b, c\}$)**.** Let $X = \{a, b, c\}$ with $q(a, b) = 1$, $q(b, a) = 3$, $q(a, c) = 2$, $q(c, a) = 0$, $q(b, c) = 1$, $q(c, b) = 2$, and $q(x, x) = 0$ for all x. One can verify that the triangle inequality holds. The forward topology τ_q has open ball $B_q(a, 1.5) = \{a, b\}$, while the conjugate topology τ_{q^t} has $B_{q^t}(a, 1.5) = \{a, c\}$ (since $q^t(a, c) = q(c, a) = 0 < 1.5$). Thus the two topologies differ: points that are "close" to a depend on the direction in which we measure distance. This is the hallmark of genuine asymmetry.

Remark 2.8 (Topological considerations)**.** A quasi-metric q on X generates a topology τ_q via the open balls $B_q(x, \varepsilon) := \{y \in X : q(x, y) < \varepsilon\}$. The conjugate q^t generates a potentially different topology τ_{q^t}. These two topologies coincide if and only if q is symmetric, i.e., if q is a metric. The study of these asymmetric topologies is a central theme in the work of Künzi [5, 6] and has deep connections to domain theory in computer science [1, 11].

3 The complexity quasi-metric space

We now turn to the specific quasi-metric space this paper is about. The *complexity space* was introduced by Schellekens [10] and further studied by Romaguera and Schellekens [9]. It is the topological framework in which the resource-usage functions associated with algorithms live.

3.1 Definition and basic properties

Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$. Let $\mathcal{C}$ denote the set of all functions

$$f\colon \mathbb{N} \to [1, \infty) \quad \text{with} \quad \sum_{n=1}^{\infty} 2^{-n}/f(n) < \infty.$$

Each such function represents the running-time profile of an algorithm: $f(n)$ is the time required on inputs of size n.

Remark 3.1 (Why $f(n) \geq 1$)**.** The lower bound $f(n) \geq 1$ encodes the elementary fact that any computation on an input of size n takes at least one elementary step. This is the standard convention in the Schellekens framework [10, 9]: without it, the quasi-metric $d_{\mathcal{C}}$ is not bounded — and in fact need not even take finite values — on the larger class of strictly positive functions. The summability requirement $\sum 2^{-n}/f(n) < \infty$ is automatic when f is bounded below by a positive constant, and ensures that $d_{\mathcal{C}}$ is everywhere finite.

Definition 3.2 (Complexity quasi-metric [10])**.** The *complexity quasi-metric* $d_{\mathcal{C}}\colon \mathcal{C} \times \mathcal{C} \to [0, \infty)$ is defined by

$$d_{\mathcal{C}}(f, g) \;=\; \sum_{n=1}^{\infty} 2^{-n} \max\left\{0,\; \frac{1}{g(n)} - \frac{1}{f(n)}\right\}.$$

The reciprocals $1/f(n)$ and $1/g(n)$ should be thought of as *efficiency measures*: a faster algorithm has a larger reciprocal. The difference $1/g(n) - 1/f(n)$ is positive when g is more efficient than f at input size n, and the weighting 2^{-n} ensures convergence of the series.

Theorem 3.3 (Basic properties of $d_{\mathcal{C}}$)**.** *The following hold:*

(i) *$d_{\mathcal{C}}$ is a quasi-metric on $\mathcal{C}$.*

(ii) *$d_{\mathcal{C}}(f, g) = 0$ if and only if $f(n) \leq g(n)$ for all $n \in \mathbb{N}$.*

(iii) *$d_{\mathcal{C}}(f, g) \leq 1$ for all $f, g \in \mathcal{C}$.*

Proof. **(i)** Axiom (Q1): $d_{\mathcal{C}}(f, f) = \sum 2^{-n} \max\{0, 0\} = 0$. The triangle inequality (Q2): for each n,

$$\max\left\{0, \frac{1}{h(n)} - \frac{1}{f(n)}\right\} \;\leq\; \max\left\{0, \frac{1}{g(n)} - \frac{1}{f(n)}\right\} + \max\left\{0, \frac{1}{h(n)} - \frac{1}{g(n)}\right\},$$

since the positive part is subadditive. Multiplying by 2^{-n} and summing gives $d_{\mathcal{C}}(f, h) \leq d_{\mathcal{C}}(f, g) + d_{\mathcal{C}}(g, h)$. Axiom (Q3): if $d_{\mathcal{C}}(f, g) = 0 = d_{\mathcal{C}}(g, f)$, then $f(n) \leq g(n)$ and $g(n) \leq f(n)$ for all n, so $f = g$.
(ii) $d_{\mathcal{C}}(f, g) = 0$ iff each term is zero, iff $1/g(n) \leq 1/f(n)$ for all n, iff $f(n) \leq g(n)$.
(iii) Each term $2^{-n} \max\{0,\, 1/g(n) - 1/f(n)\} \leq 2^{-n} \cdot 1/g(n) \leq 2^{-n}$, so $d_{\mathcal{C}}(f, g) \leq \sum_{n=1}^{\infty} 2^{-n} = 1$. □

Property (ii) is the key asymmetry result: *moving from a faster function to a slower one is free*, because $f(n) \leq g(n)$ (i.e., f is faster) implies $d_{\mathcal{C}}(f, g) = 0$.

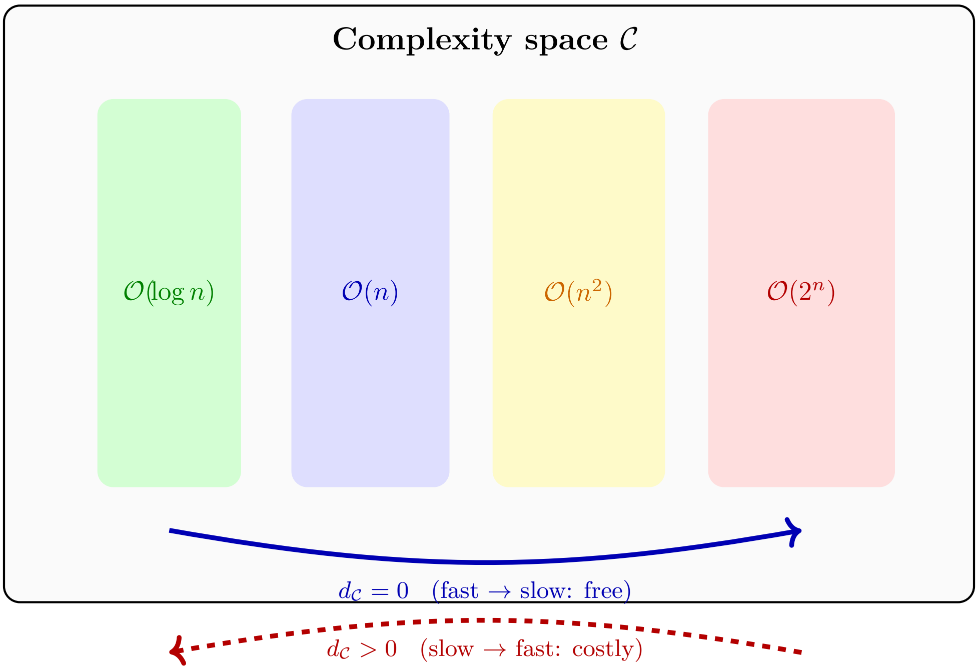

Figure 4: The complexity landscape. Moving from a faster class to a slower one is free ($d_{\mathcal{C}} = 0$); the reverse direction is costly ($d_{\mathcal{C}} > 0$).

3.2 Illustrative examples

A few worked examples make the definition concrete.

Example 3.4 (Linear vs. quadratic)**.** Let $f(n) = n$ and $g(n) = n^2$. Since $f(n) \le g(n)$ for all $n \ge 1$, Theorem 3.3(ii) gives $d_{\mathcal{C}}(f, g) = 0$. In the reverse direction,

$$d_{\mathcal{C}}(g, f) = \sum_{n=1}^{\infty} 2^{-n} \max\left\{0, \frac{1}{n} - \frac{1}{n^2}\right\} = \sum_{n=2}^{\infty} 2^{-n} \cdot \frac{n-1}{n^2},$$

since the $n = 1$ term vanishes ($\frac{1}{1} - \frac{1}{1} = 0$). We compute the first few partial sums to see how the series converges:

$$\begin{aligned}
S_2 &= \tfrac{1}{4} \cdot \tfrac{1}{4} = 0.0625,\\
S_3 &= S_2 + \tfrac{1}{8} \cdot \tfrac{2}{9} = 0.0625 + 0.0278 = 0.0903,\\
S_4 &= S_3 + \tfrac{1}{16} \cdot \tfrac{3}{16} = 0.0903 + 0.0117 = 0.1020,\\
S_5 &= S_4 + \tfrac{1}{32} \cdot \tfrac{4}{25} = 0.1020 + 0.0050 = 0.1070.
\end{aligned}$$

The series converges rapidly due to the 2^{-n} factor; by $n = 10$ the partial sum is already 0.1108, within 0.001 of the limit. Analytically, splitting $\frac{n-1}{n^2} = \frac{1}{n} - \frac{1}{n^2}$ and using $\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1 - x)$ and $\sum_{n=1}^{\infty} \frac{x^n}{n^2} = \mathrm{Li}_2(x)$ at $x = \frac{1}{2}$ gives the closed form

$$d_{\mathcal{C}}(g, f) = \ln 2 - \mathrm{Li}_2(\tfrac{1}{2}) \approx 0.693 - 0.582 = 0.111.$$

The exact value of the series can be confirmed symbolically using SageMath; see `complexity_distances.sage` in the companion repository. The asymmetry is clear: moving from linear to quadratic is free, but moving from quadratic to linear costs approximately 0.111.

Example 3.5 (Logarithmic vs. linear)**.** Let $f(n) = \ln(n+1)$ and $g(n) = n$. Since $\ln(n+1) \le n$ for all $n \ge 1$, we have $d_{\mathcal{C}}(f,g) = 0$. But $d_{\mathcal{C}}(g,f) > 0$ because g is slower. The partial sums are:

$$S_1 = \tfrac{1}{2}\big(\tfrac{1}{\ln 2} - 1\big) \approx 0.2213, \quad S_2 = S_1 + \tfrac{1}{4}\big(\tfrac{1}{\ln 3} - \tfrac{1}{2}\big) \approx 0.3179,$$
$$S_3 \approx 0.3609, \quad S_5 \approx 0.3991, \quad S_{10} \approx 0.4165.$$

The limit is $d_{\mathcal{C}}(g,f) \approx 0.417$; for exact symbolic evaluation see `complexity_distances.sage`.

Example 3.6 (Equal functions)**.** If $f = g$, then $d_{\mathcal{C}}(f,g) = d_{\mathcal{C}}(g,f) = 0$. The distance is symmetric (and zero) for identical functions, as expected.

Example 3.7 (Constant shift)**.** Let $f(n) = n$ and $g(n) = n + c$ for some constant $c > 0$. Then $f(n) < g(n)$ for all n, so $d_{\mathcal{C}}(f,g) = 0$. In the reverse direction, $d_{\mathcal{C}}(g,f) = \sum_{n=1}^{\infty} 2^{-n} \cdot c/(n(n+c))$, which is small but positive—reflecting the fact that g is only slightly slower.

Example 3.8 (Incomparable functions)**.** Let $f(n) = n + (-1)^{n+1}$ and $g(n) = n$. Then $f(1) = 2 > 1 = g(1)$ but $f(2) = 1 < 2 = g(2)$, so neither $f(n) \le g(n)$ nor $g(n) \le f(n)$ for all n. (Note that f alternates above and below g: f exceeds g at odd indices and falls below at even indices.) Consequently, *both* $d_{\mathcal{C}}(f,g) > 0$ and $d_{\mathcal{C}}(g,f) > 0$. Only odd-index terms contribute to $d_{\mathcal{C}}(f,g)$:

$$d_{\mathcal{C}}(f,g) = \sum_{\substack{n \ge 1 \\ n \text{ odd}}} \frac{2^{-n}}{n(n+1)} = \tfrac{1}{2} \cdot \tfrac{1}{2} + \tfrac{1}{8} \cdot \tfrac{1}{12} + \cdots \approx 0.262.$$

Similarly, only even-index terms contribute to $d_{\mathcal{C}}(g,f)$:

$$d_{\mathcal{C}}(g,f) = \sum_{\substack{n \ge 2 \\ n \text{ even}}} \frac{2^{-n}}{n(n-1)} = \tfrac{1}{4} \cdot \tfrac{1}{2} + \tfrac{1}{16} \cdot \tfrac{1}{12} + \cdots \approx 0.131.$$

The symmetrized distance is $d^s_{\mathcal{C}}(f,g) = \max\{0.262, 0.131\} \approx 0.262$. Observe that $d_{\mathcal{C}}(f,g) \approx 2\, d_{\mathcal{C}}(g,f)$: the "upward oscillation" is costlier than the "downward" one, reflecting the asymmetry of the quasi-metric. The exact sums are verified symbolically in `incomparable_functions.sage`.

Example 3.9 (The bound $d_{\mathcal{C}} \le 1$ is sharp)**.** The bound in Theorem 3.3(iii) cannot be improved. Take $g(n) = 1$ for all n and, for each $k \in \mathbb{N}$, the constant function $f_k(n) = k$. Both lie in $\mathcal{C}$. Since $g \le f_k$ pointwise, we have $d_{\mathcal{C}}(g, f_k) = 0$. In the reverse direction,

$$d_{\mathcal{C}}(f_k, g) \;=\; \sum_{n=1}^{\infty} 2^{-n} \left(1 - \tfrac{1}{k}\right) \;=\; 1 - \tfrac{1}{k},$$

which approaches 1 as $k \to \infty$. Thus $\sup_{f,g \in \mathcal{C}} d_{\mathcal{C}}(f,g) = 1$, but the value 1 itself is not attained on $\mathcal{C}$.

3.3 The conjugate and symmetrized complexity distances

Since $d_{\mathcal{C}}$ is a quasi-metric, we automatically obtain the conjugate and symmetrization.

Proposition 3.10. *The conjugate of the complexity quasi-metric is*

$$d^t_{\mathcal{C}}(f,g) \;=\; d_{\mathcal{C}}(g,f) \;=\; \sum_{n=1}^{\infty} 2^{-n} \max\left\{0, \frac{1}{f(n)} - \frac{1}{g(n)}\right\}.$$

The symmetrization is $d^s_{\mathcal{C}}(f,g) = \max\{d_{\mathcal{C}}(f,g), d_{\mathcal{C}}(g,f)\}$.

Proof. By Definition 2.2, $d^t_{\mathcal{C}}(f,g) = d_{\mathcal{C}}(g,f)$. Substituting g for the first argument and f for the second in Definition 3.2 yields the stated formula. That $d^s_{\mathcal{C}} = \max\{d_{\mathcal{C}}, d^t_{\mathcal{C}}\}$ is a metric follows from the general theory (Definition 2.2). □

The symmetrized distance $d^s_{\mathcal{C}}$ is a genuine metric on $\mathcal{C}$ and measures the "worst-case directional cost" between two complexity functions.

Example 3.11 (Symmetrization of linear vs. quadratic)**.** From Example 3.4, $d_{\mathcal{C}}(f,g) = 0$ and $d_{\mathcal{C}}(g,f) \approx 0.111$, so $d^s_{\mathcal{C}}(f,g) \approx 0.111$.

3.4 Computing the complexity quasi-metric

The following algorithm approximates $d_{\mathcal{C}}(f,g)$ numerically. Since the series has infinitely many terms, we truncate at N terms; the exponential decay of 2^{-n} guarantees rapid convergence.

Algorithm 1: Compute $d_{\mathcal{C}}(f,g)$

Input: Functions f, g; truncation parameter N
Output: Approximation of $d_{\mathcal{C}}(f,g)$

$S \leftarrow 0$
for $n \leftarrow 1$ **to** N **do**
  $\Delta \leftarrow 1/g(n) - 1/f(n)$
  if $\Delta > 0$ **then**
    $S \leftarrow S + 2^{-n}\,\Delta$
  end
end
return S

Remark 3.12 (Convergence rate)**.** The truncation error after N terms is at most 2^{-N}, since each omitted term contributes at most 2^{-n}. In practice, $N = 80$ gives accuracy well beyond double-precision floating-point.

A Python implementation of Algorithm 1 is provided in `complexity_distance.py`. For many common function pairs, the infinite series admits a closed-form evaluation via symbolic algebra; we use SageMath for exact verification throughout (see `code/sagemath/`).

4 Expansive homeomorphisms

The dynamical side of the theory comes next. An expansive homeomorphism is, roughly, a map that "spreads things out": no two distinct points stay close forever under iteration.

4.1 Definition and motivation

In a standard metric space, expansiveness was introduced by Utz [12]: a homeomorphism $\psi\colon X \to X$ is *expansive* if there exists a constant $\delta > 0$ such that for every pair of distinct points $x \neq y$, some iterate ψ^n separates them by more than δ. The constant δ is called the *expansive constant*.

Olela Otafudu et al. [7] extended this to quasi-metric spaces, where the asymmetry of the distance introduces new subtleties.

Definition 4.1 (q-expansive homeomorphism [7])**.** Let (X, q) be a T_0 quasi-metric space and let $\psi\colon X \to X$ be a homeomorphism with respect to the metric topology τ_{q^s} generated by the symmetrization $q^s = \max\{q, q^t\}$. We say that ψ is *q-expansive* if there exists a constant $\delta > 0$ (an *expansive constant*) such that for every pair of distinct points $x \neq y$ in X, there exists $n \in \mathbb{Z}$ with

$$\max\{\, q(\psi^n(x), \psi^n(y)),\ q(\psi^n(y), \psi^n(x))\,\} \ >\ \delta.$$

Note that we allow both positive and negative iterates: the separation may occur in the future or in the past. This is essential in the quasi-metric setting, because the asymmetry of q means that forward and backward iterates may behave very differently.

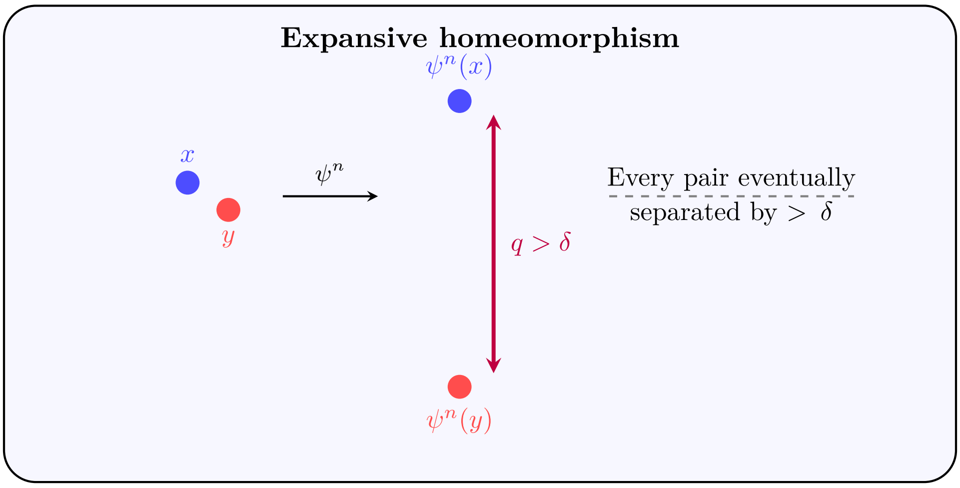

Figure 5: An expansive homeomorphism: the orbits of any two distinct points are eventually separated by more than δ.

4.2 Equivalences in the quasi-metric setting

A key result of [7] is that expansiveness with respect to q is equivalent to expansiveness with respect to the conjugate q^t.

Theorem 4.2 ([7])**.** *Let (X, q) be a quasi-metric space and $\psi\colon X \to X$ a homeomorphism. Then:*

(i) *ψ is q-expansive if and only if ψ is q^t-expansive.*

(ii) *If ψ is q-expansive, then ψ is q^s-expansive. The converse is false in general.*

Part (i) is striking: the direction of asymmetry does not matter for whether expansive behaviour *exists*, though it may change the expansive constant. Part (ii) says that quasi-metric expansiveness is strictly stronger than metric expansiveness.

Example 4.3 (Non-equivalence of (ii))**.** Consider $X = \{0, 1\}^{\mathbb{Z}}$ (the full two-shift) with the quasi-metric $q(x, y) = \sum_{n \geq 0} 2^{-n}|x_n - y_n|$ (only non-negative indices). The shift map is q^s-expansive but not q-expansive, because forward-only distances cannot detect differences in the past.

Example 4.4 (Illustrating part (i): q-expansive $\Leftrightarrow$ q^t-expansive)**.** Consider $f(n) = n$ and $g(n) = 2n$ on $(\mathcal{C}, d_{\mathcal{C}})$ with ψ_2. We have $d_{\mathcal{C}}(f, g) = 0$ (since $n \leq 2n$) and $d_{\mathcal{C}}(g, f) = \sum_{n=1}^{\infty} 2^{-n} \cdot \frac{1}{2n} = \frac{1}{2}\sum_{n=1}^{\infty} \frac{(1/2)^n}{n} = \frac{1}{2}\ln 2 \approx 0.347$ (partial sums: $S_1 = 0.250$, $S_2 = 0.313$, $S_3 = 0.333$, $S_5 = 0.344$; exact value: $\frac{1}{2}\ln 2$). The backward iterates give $d_{\mathcal{C}}(\psi_2^{-k}(g), \psi_2^{-k}(f)) = 2^k \cdot 0.347$, which exceeds any δ for k large enough. Thus ψ_2 is $d_{\mathcal{C}}$-expansive. For the conjugate: $d^t_{\mathcal{C}}(f, g) = d_{\mathcal{C}}(g, f) \approx 0.347 > 0$, and $d^t_{\mathcal{C}}(\psi_2^{-k}(f), \psi_2^{-k}(g)) = 2^k \cdot 0.347 \to \infty$. So ψ_2 is also $d^t_{\mathcal{C}}$-expansive, confirming part (i) of Theorem 4.2.

Example 4.5 (The identity is not expansive)**.** The identity map id: $\mathcal{C} \to \mathcal{C}$ is trivially a $\tau_{d^s_{\mathcal{C}}}$-homeomorphism. For any $\delta > 0$, choose $f, g \in \mathcal{C}$ with $0 < d^s_{\mathcal{C}}(f, g) < \delta$ (for example, two distinct constant functions $f \equiv k_1$, $g \equiv k_2$ with $|k_1 - k_2|$ small). Every iterate of id leaves the distance unchanged, so $d^s_{\mathcal{C}}(\mathrm{id}^n(f), \mathrm{id}^n(g)) < \delta$ for all $n \in \mathbb{Z}$. Hence id is not expansive. This example shows that the conclusion of Theorem 5.9 can fail even for $\tau_{d^s_{\mathcal{C}}}$-homeomorphisms when $\alpha = 1$.

4.3 Checking expansiveness numerically

Given two functions f, g and a candidate map ψ, we can numerically check whether their orbits separate. The idea is simple: iterate ψ both forward and backward and check whether the distance exceeds δ at some iterate.

Algorithm 2: Check expansive separation

Input: Functions $f \neq g$; map ψ; candidate δ; iteration bound M
Output: `True` if separation $> \delta$ found; the iterate n

```
for n ← −M to M do
    d ← q(ψ^n(f), ψ^n(g))
    if d > δ then
        return True, n
    end
end
return False, None
```

A Python implementation is given in `expansiveness_check.py`.

5 The scaling transformation

The main dynamical object of the paper is the *scaling transformation* on the complexity space—the simplest non-trivial map that respects the multiplicative structure of running-time functions.

5.1 Definition and basic properties

Definition 5.1 (Scaling transformation)**.** For $\alpha > 0$, the *scaling transformation* $\psi_\alpha : \mathcal{C} \to \mathcal{C}$ is defined by

$$\psi_\alpha(f)(n) \;=\; \alpha \cdot f(n).$$

Its k-th iterate is $\psi^k_\alpha(f)(n) = \alpha^k f(n)$.

The map ψ_α multiplies every running-time value by the constant α. When $\alpha > 1$, this makes algorithms "slower" (larger running times); when $0 < \alpha < 1$, it makes them "faster." When $\alpha = 1$, it is the identity.

Example 5.2 (Scaling a linear function)**.** If $f(n) = n$ and $\alpha = 2$, then $\psi_2(f)(n) = 2n$, $\psi^2_2(f)(n) = 4n$, $\psi^3_2(f)(n) = 8n$, and in general $\psi^k_2(f)(n) = 2^k n$. The orbit of f under ψ_2 consists of all functions of the form $2^k n$ for $k \in \mathbb{Z}$.

Example 5.3 (Scaling a quadratic function)**.** If $g(n) = n^2$ and $\alpha = 3$, then $\psi^k_3(g)(n) = 3^k n^2$. The orbit consists of functions $3^k n^2$, which are all quadratic but with different leading coefficients.

The key algebraic property of ψ_α with respect to $d_{\mathcal{C}}$ is that it acts as a *Lipschitz contraction* (when $\alpha > 1$) or *expansion* (when $\alpha < 1$).

Lemma 5.4. *For any $\alpha > 0$ and any $f, g \in \mathcal{C}$,*

$$d_{\mathcal{C}}(\psi_\alpha(f), \psi_\alpha(g)) \;=\; \frac{1}{\alpha}\, d_{\mathcal{C}}(f, g).$$

Proof. We compute directly:

$$\begin{aligned} d_{\mathcal{C}}(\psi_\alpha(f), \psi_\alpha(g)) &= \sum_{n=1}^{\infty} 2^{-n} \max\Big\{0, \frac{1}{\alpha g(n)} - \frac{1}{\alpha f(n)}\Big\} \\ &= \sum_{n=1}^{\infty} 2^{-n} \cdot \frac{1}{\alpha} \max\Big\{0, \frac{1}{g(n)} - \frac{1}{f(n)}\Big\} \;=\; \frac{1}{\alpha}\, d_{\mathcal{C}}(f, g). \end{aligned}$$

□

By induction, $d_{\mathcal{C}}(\psi_\alpha^k(f), \psi_\alpha^k(g)) = \alpha^{-k}\, d_{\mathcal{C}}(f, g)$ for all $k \geq 0$.

Example 5.5 (Numerical verification of Lemma 5.4)**.** Let $f(n) = n$, $g(n) = n^2$, and $\alpha = 3$. We have $d_{\mathcal{C}}(g, f) \approx 0.111$. After scaling: $\psi_3(f)(n) = 3n$ and $\psi_3(g)(n) = 3n^2$. Then $d_{\mathcal{C}}(\psi_3(g), \psi_3(f)) = d_{\mathcal{C}}(3n^2, 3n) = \sum_{n=1}^{\infty} 2^{-n} \max\Big\{0, \frac{1}{3n} - \frac{1}{3n^2}\Big\} = \frac{1}{3} \cdot d_{\mathcal{C}}(g, f) \approx 0.037$, which matches $\frac{1}{\alpha} \cdot 0.111 = 0.037$ exactly as confirmed by SageMath (`partial_sums.sage`).

Remark 5.6 (Group structure)**.** The scaling maps form a multiplicative group: $\psi_\alpha \circ \psi_\beta = \psi_{\alpha\beta}$ and $\psi_\alpha^{-1} = \psi_{1/\alpha}$. In particular, the family $\{\psi_\alpha : \alpha > 0\}$ is isomorphic to $(\mathbb{R}_{>0}, \cdot)$. The Lipschitz property of Lemma 5.4 extends to this group action: $d_{\mathcal{C}}(\psi_\alpha \circ \psi_\beta(f), \psi_\alpha \circ \psi_\beta(g)) = \frac{1}{\alpha\beta}\, d_{\mathcal{C}}(f, g)$.

Remark 5.7 (Interpretation)**.** When $\alpha > 1$, forward iterates of ψ_α bring functions *closer* in the $d_{\mathcal{C}}$ distance (contraction by factor $1/\alpha$ per step). Backward iterates push them *apart* (expansion by factor α per step). When $0 < \alpha < 1$, the roles reverse.

5.2 Main theorem: Expansiveness of scaling

The main result says that the scaling map is expansive exactly when it is non-trivial.

Lemma 5.8 (ψ_α is a homeomorphism)**.** *For every $\alpha > 0$, the map $\psi_\alpha\colon \mathcal{C} \to \mathcal{C}$ is a bijection with inverse $\psi_{1/\alpha}$, and both ψ_α and $\psi_{1/\alpha}$ are Lipschitz with respect to $d^s_{\mathcal{C}}$. In particular, ψ_α is a $\tau_{d^s_{\mathcal{C}}}$-homeomorphism.*

Proof. The inverse identity $\psi_\alpha \circ \psi_{1/\alpha} = \mathrm{id}$ is immediate. By Lemma 5.4 applied to both $d_{\mathcal{C}}$ and $d^t_{\mathcal{C}}$, $d^s_{\mathcal{C}}(\psi_\alpha(f), \psi_\alpha(g)) = (1/\alpha) d^s_{\mathcal{C}}(f, g)$, giving the Lipschitz bound with constant $1/\alpha$; the same applies to $\psi_{1/\alpha}$ with constant α. Both maps are therefore $\tau_{d^s_{\mathcal{C}}}$-continuous, and ψ_α is a homeomorphism. □

Theorem 5.9 (Main theorem)**.** *The scaling transformation ψ_α is expansive on $(\mathcal{C}, d_{\mathcal{C}})$ if and only if $\alpha \neq 1$.*

Proof. We consider three cases.
Case 1: $\alpha = 1$**.** The map ψ_1 is the identity, so $d_{\mathcal{C}}(\psi_1^n(f), \psi_1^n(g)) = d_{\mathcal{C}}(f, g)$ for all n. If $d_{\mathcal{C}}(f, g) > 0$, the distance is constant and may be smaller than any proposed δ; if $d_{\mathcal{C}}(f, g) = 0$ but $d_{\mathcal{C}}(g, f) > 0$ (i.e., f is faster than g), then $d_{\mathcal{C}}(\psi_1^n(f), \psi_1^n(g)) = 0$ for all n and no separation occurs. Hence ψ_1 is not expansive.
Case 2: $\alpha > 1$. Take any $\delta \in (0, 1)$. Let $f \neq g$ in $\mathcal{C}$. Since $(\mathcal{C}, d^s_{\mathcal{C}})$ is a T_0 quasi-metric space, at

least one of $d_{\mathcal{C}}(f,g)$ or $d_{\mathcal{C}}(g,f)$ is positive.
If $d_{\mathcal{C}}(f,g) > 0$, apply Corollary 7.7 with the pair (f,g): for some $k \geq 0$, $d_{\mathcal{C}}(\psi_\alpha^{-k}(f), \psi_\alpha^{-k}(g)) = \alpha^k d_{\mathcal{C}}(f,g) > \delta$, and Definition 4.1 is satisfied with $n = -k$.
If instead $d_{\mathcal{C}}(f,g) = 0$ but $d_{\mathcal{C}}(g,f) > 0$, apply the same corollary to the pair (g,f): for some $k \geq 0$, $d_{\mathcal{C}}(\psi_\alpha^{-k}(g), \psi_\alpha^{-k}(f)) = \alpha^k d_{\mathcal{C}}(g,f) > \delta$. Since $d_{\mathcal{C}}(\psi_\alpha^{-k}(g), \psi_\alpha^{-k}(f)) = d_{\mathcal{C}}^t(\psi_\alpha^{-k}(f), \psi_\alpha^{-k}(g))$, the symmetric condition in Definition 4.1 (with $q = d_{\mathcal{C}}$, $n = -k$) is satisfied.
Case 3: $0 < \alpha < 1$. Take any $\delta \in (0,1)$. Let $f \neq g$ in $\mathcal{C}$. At least one of $d_{\mathcal{C}}(f,g)$ or $d_{\mathcal{C}}(g,f)$ is positive.
If $d_{\mathcal{C}}(f,g) > 0$, then for forward iterates: $d_{\mathcal{C}}(\psi_\alpha^k(f), \psi_\alpha^k(g)) = \alpha^{-k} d_{\mathcal{C}}(f,g) > \delta$ for k large enough. Definition 4.1 is satisfied with $n = k$.
If instead $d_{\mathcal{C}}(f,g) = 0$ but $d_{\mathcal{C}}(g,f) > 0$, apply the same argument to the pair (g,f): $d_{\mathcal{C}}(\psi_\alpha^k(g), \psi_\alpha^k(f)) = \alpha^{-k} d_{\mathcal{C}}(g,f) > \delta$ for k large, and the symmetric condition in Definition 4.1 is satisfied with $n = k$. □

Example 5.10 (Expansiveness for $\alpha = 2$)**.** Consider $f(n) = n$ and $g(n) = n+1$. For $\alpha = 2$, we have $d_{\mathcal{C}}(f,g) = 0$ (since $f(n) < g(n)$ for all n) but

$$d_{\mathcal{C}}(g,f) = \sum_{n=1}^{\infty} \frac{2^{-n}}{n(n+1)}.$$

Computing partial sums: $S_1 = \frac{1}{4} = 0.250$, $S_2 = S_1 + \frac{1}{24} \approx 0.292$, $S_3 \approx 0.302$, $S_5 \approx 0.306$, converging to ≈ 0.307 (see `complexity_distances.sage`). The backward iterates give:

$$d_{\mathcal{C}}(\psi_2^{-k}(g), \psi_2^{-k}(f)) = 2^k \cdot 0.307$$

For $\delta = 0.5$, we need $2^k \cdot 0.307 > 0.5$, i.e., $k \geq 1$. Indeed, $d_{\mathcal{C}}(\psi_2^{-1}(g), \psi_2^{-1}(f)) \approx 0.614 > 0.5$. Thus ψ_2 is expansive with expansive constant $\delta = 0.5$.

Example 5.11 (Non-expansiveness for $\alpha = 1$)**.** Take $f(n) = n$ and $g(n) = n^2$. For $\alpha = 1$, ψ_1 is the identity, so $d_{\mathcal{C}}(\psi_1^n(f), \psi_1^n(g)) = d_{\mathcal{C}}(f,g) = 0$ for all n, while $d_{\mathcal{C}}(\psi_1^n(g), \psi_1^n(f)) = d_{\mathcal{C}}(g,f) \approx 0.111$ for all n. No matter how large n is, $d_{\mathcal{C}}(f,g)$ remains 0, so no separation occurs in that direction. Therefore ψ_1 is not expansive.

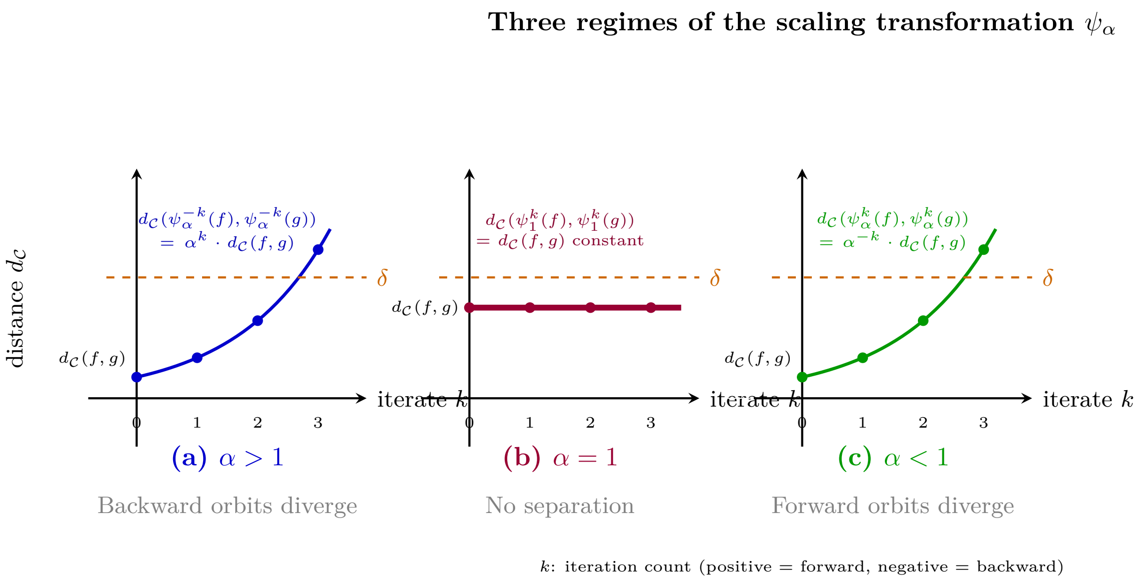

Figure 6: The three regimes of the scaling transformation ψ_α. (a) For $\alpha > 1$, backward iterates cause exponential separation: $d_\mathcal{C}(\psi_\alpha^{-k}(f), \psi_\alpha^{-k}(g)) = \alpha^k d_\mathcal{C}(f,g)$. (b) For $\alpha = 1$, the map is the identity and distances remain constant. (c) For $0 < \alpha < 1$, forward iterates cause exponential separation: $d_\mathcal{C}(\psi_\alpha^k(f), \psi_\alpha^k(g)) = \alpha^{-k} d_\mathcal{C}(f,g)$. In cases (a) and (c), for any $\delta > 0$, there exists k such that the distance exceeds δ, making ψ_α expansive.

Example 5.12 (Concrete separation for $\alpha = 2$)**.** Let $f(n) = n$, $g(n) = n+1$, and $\alpha = 2$. Then $d_\mathcal{C}(f,g) = 0$ (since $f(n) < g(n)$) and $d_\mathcal{C}(g,f) > 0$. After k backward iterates, $d_\mathcal{C}(\psi_2^{-k}(g), \psi_2^{-k}(f)) = 2^k\, d_\mathcal{C}(g,f)$. Even with $d_\mathcal{C}(g,f)$ small, a few backward steps suffice to exceed any given δ. A numerical verification is given in `expansiveness_check.py`.

Example 5.13 (Separation for $\alpha = 1/3$)**.** Let $f(n) = n^2$, $g(n) = n^3$, and $\alpha = 1/3$. Here $d_\mathcal{C}(f,g) = 0$ but $d_\mathcal{C}(g,f) > 0$. After k forward iterates, $d_\mathcal{C}(\psi_{1/3}^k(g), \psi_{1/3}^k(f)) = 3^k\, d_\mathcal{C}(g,f) \to \infty$.

5.3 The expansive constant

For a given $\alpha \neq 1$ and pair $f \neq g$, one may ask: what is the *smallest* iterate n at which separation occurs? This depends on the initial distance and the value of α.

Proposition 5.14 (Separation iterate)**.** *Let $\alpha > 1$ and let $f, g \in \mathcal{C}$ with $d := d_\mathcal{C}(f,g) > 0$. Then $d_\mathcal{C}(\psi_\alpha^{-k}(f), \psi_\alpha^{-k}(g)) > \delta$ for all $k \geq \lceil \log_\alpha(\delta/d) \rceil$.*

Proof. We need $\alpha^k d > \delta$, i.e., $k > \log_\alpha(\delta/d)$. □

Example 5.15 (Computing the separation iterate)**.** Let $\alpha = 2$, $f(n) = n^2$, $g(n) = n$, and $\delta = 0.5$. Since $f(n) \geq g(n)$ for $n \geq 1$, we have $d := d_\mathcal{C}(f,g) > 0$. Numerically, $d \approx 0.111$. By Proposition 5.14, separation occurs for $k \geq \lceil \log_2(0.5/0.111) \rceil = \lceil \log_2(4.50) \rceil = \lceil 2.17 \rceil = 3$. Indeed, $d_\mathcal{C}(\psi_2^{-3}(f), \psi_2^{-3}(g)) = 8 \cdot 0.111 = 0.888 > 0.5$. For $k = 2$: $4 \cdot 0.111 = 0.444 < 0.5$, confirming that $k = 3$ is the *first* iterate achieving separation (verified in `separation_iterates.sage`).

Example 5.16 (Counterexample: α close to 1 delays separation)**.** Let $\alpha = 1.01$, $f(n) = n$, $g(n) = n+1$, and $\delta = 0.5$. Then $d := d_\mathcal{C}(g,f) \approx 0.307$. The separation iterate satisfies $k \geq \lceil \log_{1.01}(0.5/0.307) \rceil = \lceil \log_{1.01}(1.63) \rceil \approx \lceil 49.1 \rceil = 50$. Thus α close to 1 requires 50 backward iterates for separation, while $\alpha = 2$ needs only 3 (Example 5.15). This illustrates that the "speed" of expansiveness is controlled by $\log \alpha$ (see `separation_iterates.sage`).

6 Stable and unstable sets

After expansiveness comes the study of *stable* and *unstable sets*: the points whose orbits stay close to a given orbit forward or backward in time. In our setting these sets have a clean interpretation in terms of complexity classes.

6.1 Stable sets

Definition 6.1 (Stable set)**.** Let (X, q) be a quasi-metric space and $\psi\colon X \to X$ a homeomorphism. The δ-*stable set* of $f \in X$ is

$$S_q(f, \delta, \psi) \;=\; \{g \in X : q(\psi^n(f), \psi^n(g)) \leq \delta \text{ for all } n \geq 0\}.$$

For the scaling transformation on the complexity space, the stable sets are easy to describe.

Theorem 6.2 (Stable sets = complexity classes)**.** *For $\alpha > 1$, the δ-stable set of f under ψ_α in the complexity quasi-metric is*

$$S_{d_{\mathcal{C}}}(f, \delta, \psi_\alpha) \;=\; \{g \in \mathcal{C} : d_{\mathcal{C}}(f, g) \leq \delta\}.$$

In particular, this set contains all g with $g(n) \geq f(n)$ for every n—that is, all functions that are "at least as slow as f."

Proof. For $n \geq 0$, $d_{\mathcal{C}}(\psi_\alpha^n(f), \psi_\alpha^n(g)) = \alpha^{-n} d_{\mathcal{C}}(f, g)$. Since $\alpha > 1$, this is a decreasing sequence, maximized at $n = 0$ where it equals $d_{\mathcal{C}}(f, g)$. Therefore

$$g \in S_{d_{\mathcal{C}}}(f, \delta, \psi_\alpha) \iff \sup_{n \geq 0} \alpha^{-n} d_{\mathcal{C}}(f, g) \leq \delta \iff d_{\mathcal{C}}(f, g) \leq \delta.$$

If $g(n) \geq f(n)$ for all n, then $d_{\mathcal{C}}(f, g) = 0 \leq \delta$, so g is in the stable set. □

Example 6.3 (Stable set of $f(n) = n$)**.** Take $f(n) = n$, $\alpha = 2$, $\delta = 0.1$. The stable set includes:

- $g_1(n) = n^2$ ($d_{\mathcal{C}}(f, g_1) = 0 \leq 0.1$)
- $g_2(n) = 2n$ ($d_{\mathcal{C}}(f, g_2) = 0 \leq 0.1$)
- $g_3(n) = n + 10$ ($d_{\mathcal{C}}(f, g_3) = 0 \leq 0.1$)
- $g_4(n) = n\sqrt{n}$ ($d_{\mathcal{C}}(f, g_4) = 0 \leq 0.1$, since $n \leq n\sqrt{n}$)

However, $h(n) = \sqrt{n}$ is NOT in the stable set because $d_{\mathcal{C}}(f, h) \approx 0.113 > 0.1$.

Example 6.4 (Counterexample: stability depends on δ)**.** For $f(n) = n$, $g(n) = \sqrt{n}$, and $\alpha = 2$, we have $d_{\mathcal{C}}(f, g) = \sum_{n=2}^{\infty} 2^{-n}(\frac{1}{\sqrt{n}} - \frac{1}{n}) \approx 0.113$ (since $\sqrt{n} < n$ for $n \geq 2$).

- If $\delta = 0.2$, then $g \in S_{d_{\mathcal{C}}}(f, 0.2, \psi_2)$ since $d_{\mathcal{C}}(f, g) \approx 0.113 < 0.2$.
- If $\delta = 0.05$, then $g \notin S_{d_{\mathcal{C}}}(f, 0.05, \psi_2)$ since $d_{\mathcal{C}}(f, g) \approx 0.113 > 0.05$.

This shows the δ-stable set shrinks as δ decreases (see `separation_iterates.sage` for verification).

Complexity-theoretic interpretation

The stable set $S_{d_{\mathcal{C}}}(f, \delta, \psi_\alpha)$ is the "neighbourhood of slower functions around f." In complexity theory terms, it is a kind of asymptotic complexity class: it contains all functions whose running time is "close to or worse than" that of f, where closeness is measured by $d_{\mathcal{C}}$.

6.2 Unstable sets

Definition 6.5 (Unstable set)**.** The δ-*unstable set* of f is

$$U_q(f, \delta, \psi) \;=\; \{g \in X : q(\psi^{-n}(f), \psi^{-n}(g)) \leq \delta \text{ for all } n \geq 0\}.$$

Theorem 6.6 (Unstable sets)**.** *For $\alpha > 1$, the δ-unstable set of f under ψ_α with respect to the* conjugate *quasi-metric $d^t_{\mathcal{C}}$ is*

$$U_{d^t_{\mathcal{C}}}(f, \delta, \psi_\alpha) \;=\; \{g \in \mathcal{C} : d_{\mathcal{C}}(g, f) = 0\}.$$

In particular, this set is independent of δ and equals the set of all g with $g(n) \leq f(n)$ for every n—that is, all functions that are "at least as fast as f."

Proof. By definition, $g \in U_{d^t_{\mathcal{C}}}(f, \delta, \psi_\alpha)$ iff $d^t_{\mathcal{C}}(\psi_\alpha^{-n}(f), \psi_\alpha^{-n}(g)) \leq \delta$ for all $n \geq 0$. Since $d^t_{\mathcal{C}}(h_1, h_2) = d_{\mathcal{C}}(h_2, h_1)$, this becomes $d_{\mathcal{C}}(\psi_\alpha^{-n}(g), \psi_\alpha^{-n}(f)) \leq \delta$ for all $n \geq 0$. By Lemma 5.4,

$$d_{\mathcal{C}}(\psi_\alpha^{-n}(g), \psi_\alpha^{-n}(f)) \;=\; \alpha^n \, d_{\mathcal{C}}(g, f).$$

Since $\alpha > 1$, this is an increasing sequence. If $d_{\mathcal{C}}(g, f) > 0$, then $\alpha^n d_{\mathcal{C}}(g, f) \to \infty$, which eventually exceeds δ. Hence g is in the unstable set if and only if $d_{\mathcal{C}}(g, f) = 0$.
By Theorem 3.3(ii), $d_{\mathcal{C}}(g, f) = 0$ if and only if $g(n) \leq f(n)$ for all n, so the unstable set consists precisely of the functions that are pointwise at most f. □

Example 6.7 (Unstable set of $f(n) = n^2$)**.** Take $f(n) = n^2$, $\alpha = 2$, $\delta = 0.1$. By Theorem 6.6, the unstable set consists of all g with $d_{\mathcal{C}}(g, f) = 0$, i.e., $g(n) \leq f(n) = n^2$ for all n. It includes:

- $g_1(n) = n$ ($n \leq n^2$ for all $n \geq 1$, so $d_{\mathcal{C}}(g_1, f) = 0$)
- $g_2(n) = n \log(n + 1)$ ($n \log(n + 1) \leq n^2$ for all $n \geq 1$, so $d_{\mathcal{C}}(g_2, f) = 0$)
- $g_3(n) = n^{1.5}$ ($n^{1.5} \leq n^2$ for all $n \geq 1$, so $d_{\mathcal{C}}(g_3, f) = 0$)

But $h(n) = 2^n$ is NOT in the unstable set because $2^n > n^2$ for large n, so $d_{\mathcal{C}}(h, f) > 0$.

Duality of stable and unstable sets

The stable set contains the functions *close to or slower* than f (a $d_{\mathcal{C}}$-neighbourhood); the unstable set contains exactly the functions *pointwise faster* than f (the zero-set of $d_{\mathcal{C}}(\cdot, f)$). Stable sets depend on δ; unstable sets do not. The asymmetry tracks the conjugate pair $d_{\mathcal{C}} \leftrightarrow d^t_{\mathcal{C}}$: forward iterates of ψ_α contract distances and preserve the stable neighbourhood, while backward iterates expand them and collapse the unstable set to its core.

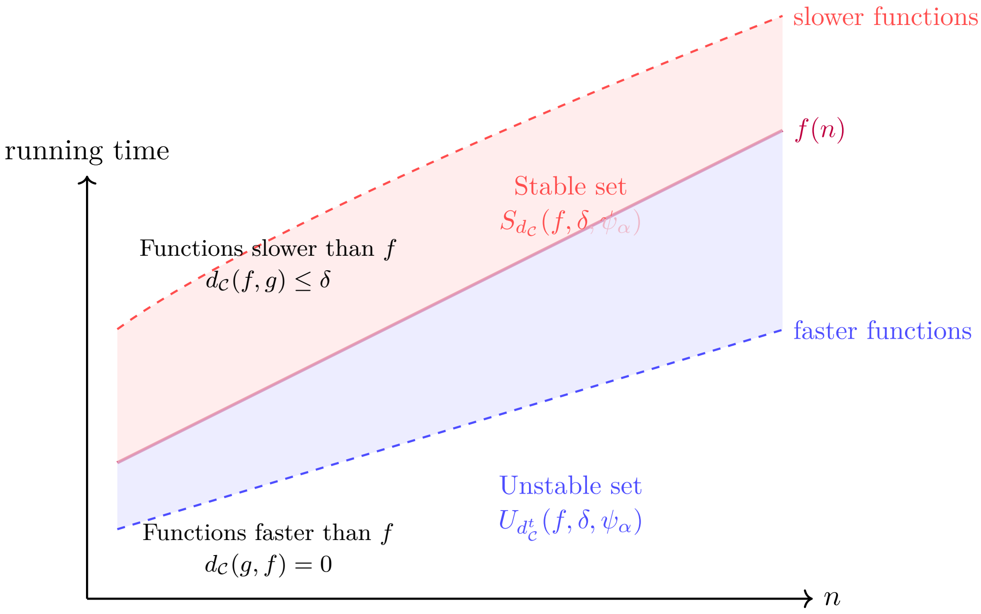

Figure 7: Stable and unstable sets for a function f. The stable set contains functions that are asymptotically slower than f; the unstable set contains functions that are asymptotically faster than f.

6.3 Algorithm for stable-set membership

The following algorithm checks whether a candidate function g belongs to the δ-stable set of f.

Algorithm 3: Membership in δ-stable set

Input: f; candidate g; α; δ; forward steps M

Output: `True` if $g \in S(f, \delta, \psi_\alpha)$

for $n \leftarrow 0$ **to** M **do**
  $s \leftarrow \alpha^n$
  $d \leftarrow d_{\mathcal{C}}(s \cdot f,\ s \cdot g)$
  if $d > \delta$ **then**
    return *False*
  end
end
return *True*

Example 6.8 (Algorithm 3 in action)**.** We trace Algorithm 3 for $f(n) = n$, $g(n) = \sqrt{n}$, $\alpha = 2$, $\delta = 0.1$, $M = 3$. At each step n, the scaled functions are $s \cdot f$ and $s \cdot g$ where $s = \alpha^n$:

n	$s = 2^n$	$d_{\mathcal{C}}(s \cdot f,\ s \cdot g)$	$d > \delta$?
0	1	$d_{\mathcal{C}}(n, \sqrt{n}) \approx 0.113$	Yes

The algorithm returns `False` at $n = 0$: $g = \sqrt{n}$ is *not* in the stable set because $d_{\mathcal{C}}(f, g) \approx 0.113 > 0.1$. In contrast, for $g(n) = 2n$: $d_{\mathcal{C}}(n, 2n) = 0 \leq 0.1$ at $n = 0$, and $d_{\mathcal{C}}(2^n \cdot n, 2^n \cdot 2n) = 0 \leq 0.1$ for all subsequent n. The algorithm returns `True`: $g = 2n$ is in the stable set.

Example 6.9 (Stable set of an exponential function)**.** Take $f(n) = 2^n$, $\alpha = 2$, $\delta = 0.01$. Since $d_{\mathcal{C}}(f, g) = 0$ whenever $g(n) \geq 2^n$ for all n, the stable set includes all super-exponential functions, such as $g(n) = 3^n$, $g(n) = n!$, and $g(n) = 2^{n^2}$. However, $h(n) = n^{100}$ is NOT in the stable set: for large n, $n^{100} < 2^n$, so $d_{\mathcal{C}}(f, h) > 0$. The numerical value $d_{\mathcal{C}}(f, h)$ is extremely close to 1 (the maximum), reflecting the vast gap between polynomial and exponential growth. This shows that even a degree-100 polynomial is far from the stable set of an exponential function.

Example 6.10 (Unstable set of a linear function)**.** Take $f(n) = n$, $\alpha = 2$, $\delta = 0.1$. By Theorem 6.6, the unstable set consists of all g with $g(n) \le n$ for all n. This includes:

- $g(n) = \log(n + 1)$ (logarithmic is faster than linear)
- $g(n) = \sqrt{n}$ (sub-linear)
- $g(n) = 1$ (constant time)
- $g(n) = n/(n + 1)$ (bounded, approaching 1)

But $h(n) = n + 1$ is NOT in the unstable set: $h(1) = 2 > 1 = f(1)$, so $d_{\mathcal{C}}(h, f) > 0$. Remarkably, adding just $+1$ to a linear function ejects it from the unstable set. This sensitivity reflects the pointwise nature of the condition $g(n) \le f(n)$ for *all* n.

Example 6.11 (Intersection of stable and unstable sets)**.** For $f(n) = n$ and $\alpha = 2$, the stable set (with $\delta > 0$) contains all g with $d_{\mathcal{C}}(f, g) \le \delta$, while the unstable set contains all g with $g(n) \le n$ for all n. A function g lies in both sets if and only if $g(n) \le n$ for all n (unstable condition) and $d_{\mathcal{C}}(f, g) \le \delta$ (stable condition). Since $g(n) \le f(n) = n$ implies $d_{\mathcal{C}}(f, g) = 0 \le \delta$, the intersection equals the unstable set itself: $S \cap U = U$. This is a general phenomenon: for $\alpha > 1$, the unstable set is always contained in every δ-stable set.

A Python implementation is given in `stable_set.py`.

7 Canonical coordinates and hyperbolicity

7.1 Background

In classical smooth dynamics, hyperbolicity is the structural property that governs much of the chaotic behaviour one wants to understand. A diffeomorphism on a compact manifold is *hyperbolic* (or *Anosov*) when the tangent bundle splits into stable and unstable sub-bundles, with the derivative contracting on one and expanding on the other. Bowen [2] showed that Anosov diffeomorphisms admit Markov partitions and satisfy strong statistical properties: equilibrium states exist and entropy formulas are exact. This is why hyperbolicity occupies such a central place in ergodic theory.

Reddy [8] later showed that these conclusions extend beyond the smooth setting: any expansive homeomorphism on a compact metric space that admits *canonical coordinates*—a local product structure in which nearby points decompose uniquely along stable and unstable directions—is hyperbolic. Reddy's theorem gives exponential contraction along one factor and exponential expansion along the other.

The complexity quasi-metric space $(\mathcal{C}, d_{\mathcal{C}})$ is not compact and $d_{\mathcal{C}}$ is not symmetric, so neither Bowen's smooth theory nor Reddy's topological generalization applies directly. However, the algebraic structure of ψ_α is explicit enough that hyperbolicity can be verified by a direct computation, and the result is in fact stronger than what the classical theory gives: we obtain *exact* geometric decay and growth, with constant $C = 1$, not just an exponential bound.

7.2 Statement and proof

Corollary 7.1 (Exact contraction rate)**.** *For every $\alpha > 0$ and every $f, g \in \mathcal{C}$,*

$$d_{\mathcal{C}}(\psi_\alpha^n(f), \psi_\alpha^n(g)) \;=\; \alpha^{-n}\, d_{\mathcal{C}}(f,g) \qquad \textit{for all } n \geq 0.$$

Proof. Induction on n, using Lemma 5.4 at each step. □

Remark 7.2 (On the term "hyperbolicity")**.** Corollary 7.1 gives an *exact* contraction rate, not merely an exponential bound; in particular, the constant in front of the exponential is 1. The terminology "hyperbolic" is often used in this kind of one-sided contraction setting (e.g., Reddy [8]), but it should be emphasized that we do *not* construct a stable/unstable splitting with a local product structure in the sense of classical hyperbolic dynamics: the dynamics here is conformal across all of $\mathcal{C}$. What we obtain is the explicit algebraic form of the contraction, which the abstract theory only delivers as an exponential bound.

Example 7.3 (Hyperbolic contraction for $\alpha = 2$)**.** Take $f(n) = n^3$, $g(n) = n^2$, $\alpha = 2$. Since $f(n) \geq g(n)$ for all $n \geq 1$, we have $d_{\mathcal{C}}(f,g) = \sum_{n=2}^{\infty} 2^{-n}\frac{n-1}{n^3} > 0$; the partial sums give $S_2 = 0.031$, $S_3 = 0.041$, $S_5 = 0.044$, converging to $d_{\mathcal{C}}(f,g) \approx 0.045$. Then:

$$\begin{aligned} d_{\mathcal{C}}(f,g) &\approx 0.045 \\ d_{\mathcal{C}}(\psi_2(f), \psi_2(g)) &= \tfrac{1}{2}\cdot 0.045 \approx 0.023 \\ d_{\mathcal{C}}(\psi_2^2(f), \psi_2^2(g)) &= \tfrac{1}{4}\cdot 0.045 \approx 0.011 \\ d_{\mathcal{C}}(\psi_2^3(f), \psi_2^3(g)) &= \tfrac{1}{8}\cdot 0.045 \approx 0.006 \end{aligned}$$

The distances contract exactly by factor 1/2 at each step; see `hyperbolic_contraction.sage` for the full sequence.

Example 7.4 (Counterexample: $\alpha = 1$ is not hyperbolic)**.** For $\alpha = 1$, ψ_1 is the identity, so:

$$d_{\mathcal{C}}(\psi_1^n(f), \psi_1^n(g)) = d_{\mathcal{C}}(f,g) \quad \text{for all } n.$$

There is no contraction or expansion—the distance remains constant. Thus ψ_1 is not hyperbolic.

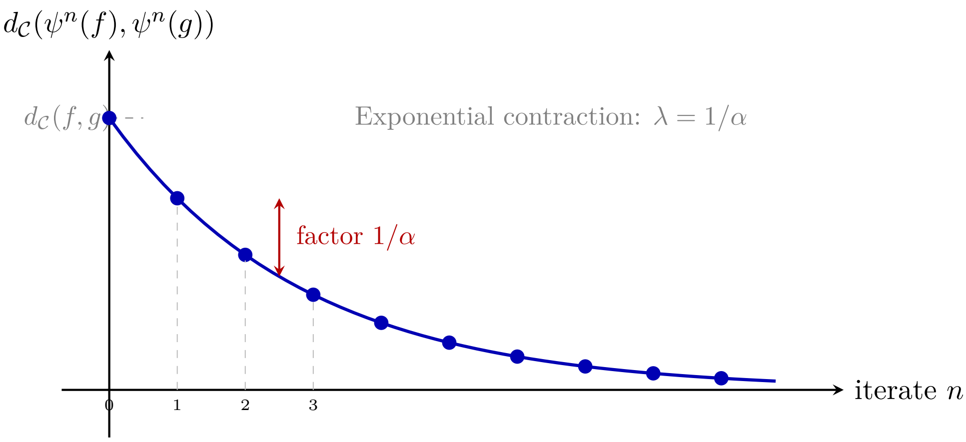

Figure 8: Exponential contraction of distances under forward iteration of ψ_α ($\alpha > 1$). The distance decays geometrically with ratio $1/\alpha$.

Remark 7.5 (Sharpness)**.** The constant $C = 1$ in Corollary 7.1 is optimal: the decay is *exactly* geometric, not merely bounded by a geometric sequence. This is a consequence of the exact scaling property of Lemma 5.4.

Example 7.6 (Numerical verification)**.** Let $f(n) = n^2$, $g(n) = n$, and $\alpha = 2$. Since $f(n) \geq g(n)$, we have $d_{\mathcal{C}}(f, g) = d_0 \approx 0.111$. After n iterates:

$$d_{\mathcal{C}}(\psi_2^n(f), \psi_2^n(g)) = 2^{-n} \cdot d_0.$$

At $n = 5$, the predicted distance is $d_0/32 \approx 0.00347$, which matches the numerical computation to full floating-point precision. See `hyperbolicity.py` and `hyperbolic_contraction.sage` for verification.

7.3 Backward iterates: expansion

Forward iterates contract; backward iterates expand:

$$d_{\mathcal{C}}(\psi_\alpha^{-n}(f), \psi_\alpha^{-n}(g)) = \alpha^n \, d_{\mathcal{C}}(f, g).$$

Contraction in one time direction and expansion in the other is the defining feature of hyperbolic dynamics.

Corollary 7.7. *For $\alpha > 1$ and $d_{\mathcal{C}}(f, g) > 0$, the backward orbit distances grow exponentially: $d_{\mathcal{C}}(\psi_\alpha^{-n}(f), \psi_\alpha^{-n}(g)) \to \infty$ as $n \to \infty$.*

Proof. Since $\psi_\alpha^{-1} = \psi_{1/\alpha}$, we have $d_{\mathcal{C}}(\psi_\alpha^{-n}(f), \psi_\alpha^{-n}(g)) = d_{\mathcal{C}}(\psi_{1/\alpha}^n(f), \psi_{1/\alpha}^n(g)) = (1/\alpha)^{-n} \, d_{\mathcal{C}}(f, g) = \alpha^n \, d_{\mathcal{C}}(f, g) \to \infty$. □

Example 7.8 (Backward expansion: numerical illustration)**.** Let $f(n) = n^2$, $g(n) = n$, and $\alpha = 3$. Then $d := d_{\mathcal{C}}(f, g) \approx 0.111$. The backward orbit distances are:

k	$d_{\mathcal{C}}(\psi_3^{-k}(f), \psi_3^{-k}(g))$	Value
0	d	0.111
1	$3d$	0.333
2	$9d$	0.999
3	$27d$	2.997
4	$81d$	8.991
5	$243d$	26.97

The values grow unboundedly: this is the dynamical signature of backward expansion, and matches the Lipschitz identity of Lemma 5.4 (or its iteration in Corollary 7.7) exactly. See `hyperbolic_contraction.sage` for the full computation.

Example 7.9 (Counterexample: no expansion when $d_{\mathcal{C}}(f, g) = 0$)**.** Let $f(n) = n$ and $g(n) = n^2$ with $\alpha = 2$. Since $f(n) \leq g(n)$ for all $n \geq 1$, we have $d_{\mathcal{C}}(f, g) = 0$. Hence $d_{\mathcal{C}}(\psi_2^{-n}(f), \psi_2^{-n}(g)) = 2^n \cdot 0 = 0$ for all n. The backward iterates produce no expansion in the $d_{\mathcal{C}}$ direction. However, $d_{\mathcal{C}}(g, f) \approx 0.111 > 0$, so the *conjugate* backward distances $d_{\mathcal{C}}^t(\psi_2^{-n}(f), \psi_2^{-n}(g)) = 2^n \cdot 0.111 \to \infty$ do expand. This asymmetry is characteristic of quasi-metric dynamics.

8 Connection to the hierarchy theorem

The *time hierarchy theorem* of Hartmanis and Stearns [4] says that more time really does buy strictly more problems. Specifically, if $f(n)\log f(n) = o(g(n))$, then $\mathrm{DTIME}(f(n)) \subsetneq \mathrm{DTIME}(g(n))$: there are problems solvable in time $g(n)$ that cannot be solved in time $f(n)$.

That classical theorem has a clean dynamical counterpart: the hierarchy gap shows up as orbit separation under the scaling transformation.

Theorem 8.1 (Hierarchy as orbit separation)**.** *Let $f, g \in \mathcal{C}$ with $d_{\mathcal{C}}(g, f) > 0$ (informally, g is not pointwise at least as fast as f). Then for every $\alpha > 0$ with $\alpha \neq 1$, the orbits $\{\psi_\alpha^n(f)\}_{n\in\mathbb{Z}}$ and $\{\psi_\alpha^n(g)\}_{n\in\mathbb{Z}}$ are eventually separated in $d_{\mathcal{C}}^s$: for every $\delta > 0$ there exists $N \in \mathbb{Z}$ with $d_{\mathcal{C}}^s(\psi_\alpha^N(f), \psi_\alpha^N(g)) > \delta$.*
In particular, the standard time-hierarchy condition $f(n)\log f(n) = o(g(n))$ [4] is a sufficient (but, as Remark 8.2 explains, not necessary) condition for $d_{\mathcal{C}}(g, f) > 0$, and hence for orbit separation.

Proof. By hypothesis $d_{\mathcal{C}}(g, f) > 0$. For $\alpha > 1$, the backward iterates give

$$d_{\mathcal{C}}(\psi_\alpha^{-k}(g), \psi_\alpha^{-k}(f)) = \alpha^k\, d_{\mathcal{C}}(g, f) \;\to\; \infty.$$

Since $d_{\mathcal{C}}^s \geq d_{\mathcal{C}}$, we have $d_{\mathcal{C}}^s(\psi_\alpha^{-k}(f), \psi_\alpha^{-k}(g)) > \delta$ for k large enough; set $N = -k$. For $0 < \alpha < 1$, the forward iterates give

$$d_{\mathcal{C}}(\psi_\alpha^{k}(g), \psi_\alpha^{k}(f)) = \alpha^{-k}\, d_{\mathcal{C}}(g, f) \;\to\; \infty,$$

and we set $N = k$.
The claim about $f(n)\log f(n) = o(g(n))$ being sufficient: that condition implies $g(n)$ dominates $f(n)$ for large n, so $1/f(n) > 1/g(n)$ eventually and $d_{\mathcal{C}}(g, f) > 0$ follows. □

Remark 8.2. The dynamical orbit-separation criterion is strictly finer than the Hartmanis–Stearns time-hierarchy criterion: any pair of functions with $d_{\mathcal{C}}(g, f) > 0$ produces orbit separation, regardless of whether they are separated by a time-hierarchy gap. Whether a quantitative orbit-separation rate corresponds in any nontrivial way to a hierarchy-style separation in DTIME is an open problem.

Example 8.3 (Linear vs. $n\log^2 n$)**.** Let $f(n) = n$ and $g(n) = n\log^2(n+1)$. Then $f(n)\log f(n) = n\log n = o(n\log^2(n+1)) = o(g(n))$, so the hierarchy condition holds. Numerically, with $\alpha = 2$ and $\delta = 0.05$, separation already occurs at iterate $k = 0$ with $d_{\mathcal{C}}^s \approx 0.541$. See `hierarchy_separation.py` and `separation_iterates.sage` for verification.

Example 8.4 (Polynomial vs. exponential)**.** Let $f(n) = n^2$ and $g(n) = 2^n$. Here $f(n)\log f(n) = 2n^2\log n = o(2^n) = o(g(n))$, so the hierarchy condition is easily satisfied. The orbit separation occurs very quickly (at $k = 0$ or $k = -1$), reflecting the huge gap between polynomial and exponential complexity.

Example 8.5 (Counterexample: insufficient gap)**.** Let $f(n) = n\log n$ and $g(n) = n\log n \cdot \log\log n$. Here $f(n)\log f(n) = n\log n \cdot \log(n\log n) \sim n\log^2 n$, while $g(n) = n\log n \cdot \log\log n$. Since $n\log^2 n$ is not $o(n\log n \cdot \log\log n)$, the hierarchy condition is NOT satisfied. Indeed, numerically $d_{\mathcal{C}}^s(f, g)$ remains small under iteration, and for small δ separation may never occur.

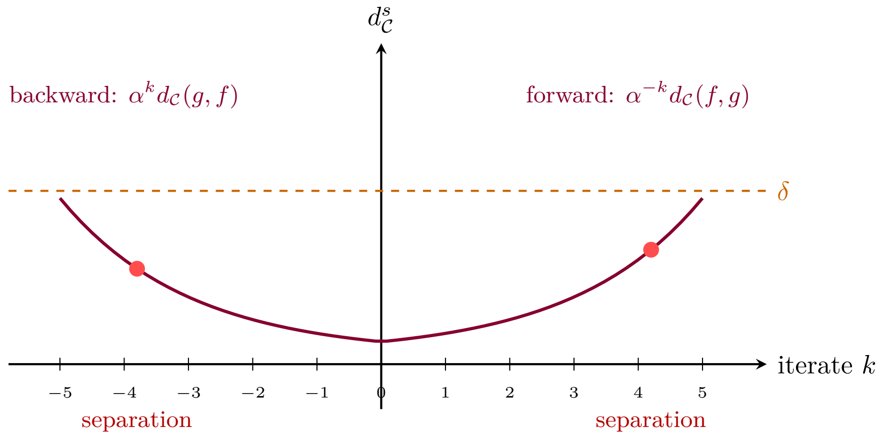

Figure 9: Orbit separation in the symmetrized metric. For functions satisfying the hierarchy gap, both forward and backward orbits eventually exceed any threshold δ.

9 Topological entropy estimates

Topological entropy is the standard invariant for measuring the "complexity of the dynamics"—the rate at which information about initial conditions is needed to predict the future. For expansive homeomorphisms on compact spaces, it is always positive [2].

This section shows that the standard compact-set entropy machinery yields only trivial information for ψ_α: every compact ψ_α-invariant subset of $(\mathcal{C}, d^s_{\mathcal{C}})$ is $d^s_{\mathcal{C}}$-trivial (Proposition 9.3). The question of a non-trivial entropy for the full non-compact dynamics is left open.

9.1 Setup and definition

Let $K \subset \mathcal{C}$ be a compact subset (in the $d^s_{\mathcal{C}}$ topology) that is ψ_α-invariant. The topological entropy $h(\psi_\alpha|_K)$ is defined via the growth rate of (n,ε)-spanning sets. A set $E \subset K$ is (n,ε)-*spanning* if for every $f \in K$ there exists $g \in E$ with $\max_{0\le j<n} d^s_{\mathcal{C}}(\psi^j_\alpha(f), \psi^j_\alpha(g)) < \varepsilon$.

Definition 9.1 (Topological entropy)**.**

$$h(\psi_\alpha|_K) = \lim_{\varepsilon\to 0} \limsup_{n\to\infty} \frac{1}{n} \log r(n,\varepsilon),$$

where $r(n,\varepsilon)$ is the minimum cardinality of an (n,ε)-spanning set.

Before stating the main entropy bound, we discuss which subsets of $\mathcal{C}$ are compact in the $d^s_{\mathcal{C}}$ topology.

Remark 9.2 (Compactness in $(\mathcal{C}, d^s_{\mathcal{C}})$)**.** The full space $\mathcal{C}$ is *not* compact in the $d^s_{\mathcal{C}}$ topology: it is unbounded (e.g., $d^s_{\mathcal{C}}(n, n^k) \to 1$ as $k \to \infty$). To obtain compact invariant sets, one must restrict attention. Two natural constructions are:

(i) *Finite orbit closures.* If K is a finite union of orbits $\{\psi^k_\alpha(f_i) : k \in \mathbb{Z}\}$, then any finite subset of K is trivially compact and ψ_α-invariant. This yields lower bounds on entropy for concrete function sets.

(ii) *Bounded complexity bands.* For $0 < a \le b$, define $K_{a,b} = \{f \in \mathcal{C} : a \le f(n) \le b \text{ for all } n\}$. Then $K_{a,b}$ is $d^s_{\mathcal{C}}$-bounded (with diameter at most 1) and closed. However, $K_{a,b}$ is ψ_α-invariant only if $a = b = 0$, which is excluded from $\mathcal{C}$. A modified band $K^N_{a,b} = \{f \in \mathcal{C} : a \le f(n) \le b \text{ for } 1 \le n \le N\}$, viewed as a subset of $\mathbb{R}^N$, is compact and can be used for finite-dimensional entropy approximations.

Proposition 9.3 below shows that any compact ψ_α-invariant subset of $(\mathcal{C}, d^s_{\mathcal{C}})$ is $d^s_{\mathcal{C}}$-trivial, so the entropy of $\psi_\alpha|_K$ is always 0 for such K.

Proposition 9.3 (No non-trivial compact invariant sets)**.** *For every $\alpha > 0$ with $\alpha \neq 1$, every non-empty compact ψ_α-invariant subset $K \subseteq (\mathcal{C}, d^s_{\mathcal{C}})$ consists of $d^s_{\mathcal{C}}$-equivalent points only: $d^s_{\mathcal{C}}(f,g) = 0$ for all $f, g \in K$.*

Proof. Suppose $\alpha > 1$ (the case $\alpha < 1$ is symmetric, replacing ψ_α by $\psi_\alpha^{-1} = \psi_{1/\alpha}$). By Corollary 7.1 applied to both $d_{\mathcal{C}}$ and its conjugate $d^t_{\mathcal{C}}$, the symmetrization satisfies

$$d^s_{\mathcal{C}}(\psi_\alpha(f), \psi_\alpha(g)) \;=\; \tfrac{1}{\alpha}\, d^s_{\mathcal{C}}(f,g),$$

so ψ_α is a strict contraction on $(\mathcal{C}, d^s_{\mathcal{C}})$ with Lipschitz constant $1/\alpha < 1$. If K is non-empty, compact, and ψ_α-invariant, then the diameter $\mathrm{diam}(K) = \sup_{f,g\in K} d^s_{\mathcal{C}}(f,g)$ satisfies $\mathrm{diam}(\psi_\alpha(K)) = (1/\alpha)\mathrm{diam}(K)$. But $\psi_\alpha(K) = K$, so $\mathrm{diam}(K) = 0$. □

Remark 9.4. Proposition 9.3 shows that the standard topological-entropy machinery (Bowen spanning sets on a compact invariant set) cannot detect any non-trivial dynamics of ψ_α on $(\mathcal{C}, d^s_{\mathcal{C}})$ directly: the only candidate compact invariant sets are $d^s_{\mathcal{C}}$-trivial. This is a consequence of the non-compactness of $\mathcal{C}$ in $\tau_{d^s_{\mathcal{C}}}$, and a known limitation of the topological-entropy framework on non-compact spaces. Alternative invariants suited to non-compact phase spaces (e.g., the volume-growth entropy of [13], or asymmetric analogues) lie outside the scope of this paper and are an open direction.

Example 9.5 (Compact invariant sets are trivial for $\alpha = 2$)**.** For $\alpha = 2$, Proposition 9.3 shows that any compact ψ_2-invariant $K \subseteq (\mathcal{C}, d^s_{\mathcal{C}})$ satisfies $d^s_{\mathcal{C}}(f,g) = 0$ for all $f, g \in K$. In particular, every two-point set $\{f, g\}$ with $d^s_{\mathcal{C}}(f,g) > 0$ (such as $f(n) = n$, $g(n) = 2n$, which have $d^s_{\mathcal{C}}(f,g) \approx 0.347 > 0$) is *not* ψ_2-invariant: ψ_2 moves it out of itself. The entropy of ψ_2 on any genuinely compact invariant set is therefore 0; non-trivial dynamical complexity of ψ_2 manifests only on non-compact invariant sets.

Example 9.6 (Singleton invariant sets)**.** Let $K = \{f\}$ for any $f \in \mathcal{C}$. Then $\psi_\alpha(K) = \{\alpha f\} \neq K$ unless $\alpha = 1$, so a non-trivial singleton is not ψ_α-invariant for $\alpha \neq 1$. This is consistent with Proposition 9.3: the only compact invariant sets are $d^s_{\mathcal{C}}$-trivial, i.e., $d^s_{\mathcal{C}}$-equivalence classes, which may be singletons only when the orbit is constant — impossible for $\alpha \neq 1$ on distinct points.

Proposition 9.7 (Finite invariant sets have zero entropy)**.** *For the scaling map ψ_α on a finite ψ_α-invariant set $K = \{f_1, \ldots, f_m\} \subset \mathcal{C}$, the topological entropy satisfies $h(\psi_\alpha|_K) = 0$.*

Proof. A finite set is a special case of a compact invariant set. By Proposition 9.3, $d^s_{\mathcal{C}}(f,g) = 0$ for all $f, g \in K$. Thus K is a single $d^s_{\mathcal{C}}$-equivalence class, the only (n, ε)-spanning set is $E = K$ itself (with $r(n,\varepsilon) = 1$ for all $n, \varepsilon > 0$), and $h(\psi_\alpha|_K) = \lim_{n\to\infty} \frac{1}{n} \log 1 = 0$. □

Remark 9.8 (Entropy on non-compact sets)**.** Proposition 9.3 shows that compact ψ_α-invariant sets carry zero entropy. The natural question is whether a meaningful entropy invariant can be defined for the non-compact dynamics of ψ_α on all of $\mathcal{C}$. On metric spaces, the variational principle equates topological entropy with the supremum of measure-theoretic entropies; an analogous quasi-metric version, if it could be established, might yield a non-trivial entropy for $\psi_\alpha|_{\mathcal{C}}$ (see Section 10).

An algorithm for numerically estimating the entropy via spanning sets is given in `entropy_estimate.py`.

10 Conclusion and open problems

This paper develops the theory of expansive homeomorphisms on the complexity quasi-metric space introduced by Schellekens. The main results are:

1. **Expansiveness characterisation** (Theorem 5.9): the scaling map ψ_α is expansive on $(\mathcal{C}, d_\mathcal{C})$ if and only if $\alpha \neq 1$.
2. **Stable sets as complexity classes** (Theorem 6.2): the δ-stable sets of ψ_α coincide with neighbourhoods in $d_\mathcal{C}$ and contain all functions that are asymptotically at least as slow.
3. **Exact contraction rate** (Corollary 7.1): distances contract exactly by factor $1/\alpha$ per forward iterate, with constant $C = 1$.
4. **Hierarchy as orbit separation** (Theorem 8.1): the time hierarchy theorem of Hartmanis and Stearns corresponds to orbit separation in the symmetrized quasi-metric.

Together, these results show that the complexity quasi-metric space is a workable setting for applying dynamical-systems methods to computational complexity.

The following table summarises the correspondence between dynamical and complexity-theoretic concepts established in this paper.

Dynamical concept	Complexity interpretation	Reference
Quasi-metric $d_\mathcal{C}(f, g) = 0$	f at least as fast as g	Thm 3.3(ii)
Scaling map ψ_α	Uniform speed change by α	Def 5.1
Expansiveness ($\alpha \neq 1$)	Orbits eventually separate	Thm 5.9
δ-stable set	Complexity class neighbourhood	Thm 6.2
Unstable set	All pointwise-faster functions	Thm 6.6
Exact contraction rate	$d_\mathcal{C}$ decays as $(1/\alpha)^n$	Cor 7.1
Backward expansion	$d_\mathcal{C}$ grows as α^n	Cor 7.7
Orbit separation	Time hierarchy gap	Thm 8.1
No non-trivial compact invariant sets	Entropy of $\psi_\alpha\|_K$ is zero	Prop 9.3

Table 1: Dictionary between dynamical systems and complexity theory.

Open problems. Several directions suggest themselves for future investigation.

Topological entropy. Proposition 9.3 shows that compact ψ_α-invariant sets are $d^s_\mathcal{C}$-trivial, so standard Bowen entropy vanishes on them. Defining a meaningful entropy invariant for the non-compact dynamics of ψ_α on all of $\mathcal{C}$ — for instance via asymmetric or volume-growth entropy analogues — remains open.

Non-linear transformations. The scaling map is linear in the function values. It would be interesting to study non-linear transformations such as $\psi(f)(n) = f(n)^2$ or $\psi(f)(n) = f(f(n))$ and determine when they are expansive.

Space complexity. The complexity quasi-metric can be adapted to space (memory) complexity. Do the results of this paper extend to that setting?

Connections to domain theory. The complexity space has deep connections to domain theory [1, 11]. It would be valuable to understand the dynamical results of this paper in domain-theoretic terms.

Stronger hierarchy results. Our Theorem 8.1 connects orbit separation to the classical time hierarchy theorem. Can stronger hierarchy-type results (e.g., the non-deterministic time hierarchy) be obtained by considering richer dynamical structures?

Composition transformations. Instead of the multiplicative scaling $\psi_\alpha(f)(n) = \alpha f(n)$, one could study composition-based transformations such as $\phi(f)(n) = f(2n)$ (input doubling) or $\phi(f)(n) = f(n^2)$ (input squaring). These maps have a different algebraic structure and would connect to speed-up theorems in complexity theory. Are they expansive on $(\mathcal{C}, d_\mathcal{C})$?

Weighted complexity quasi-metrics. Replacing the weight 2^{-n} in the definition of $d_\mathcal{C}$ by a general summable sequence $w_n > 0$ yields a family of complexity quasi-metrics. Different weight sequences emphasise different input sizes. How do the dynamical properties of ψ_α depend on the choice of weights?

Shadowing property. In classical hyperbolic dynamics, expansive homeomorphisms with the shadowing property are structurally stable. Does ψ_α on $(\mathcal{C}, d_\mathcal{C})$ possess the shadowing property? If so, this would provide a strong robustness guarantee for the dynamical characterisation of complexity classes.

Acknowledgements

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